CANE WATER AUDIT – TARGETS

Definitions, Calculation Methods, Examples and Audit Checklist

Purpose: This guide explains the target-related fields used in a cane water audit. The values entered in an audit system should reflect the plant's agreed operating targets, measurement boundaries and applicable regulatory requirements.

1. Sugar Fresh Water Target (L/t cane crushed)

Meaning: The desired quantity of fresh water used by the sugar factory for each tonne of sugarcane crushed. This is a performance target, not automatically a statutory limit.

Formula:

Sugar-house fresh water use (L/t cane) = Fresh water supplied to the defined sugar-house boundary (m³/day) × 1,000 ÷ cane crushed (t/day)

Example: If the defined sugar-house fresh water use is 750 m³/day and cane crushing is 5,000 t/day, the specific use is 750 × 1,000 ÷ 5,000 = 150 L/t cane.

How to set it: Establish a reliable baseline from calibrated flow meters, identify the boundary of the sugar house, separate internal recycled/recovered water from fresh-water intake, and set a realistic improvement target based on process requirements, equipment condition and verified performance.

2. Sugar Fresh Water Alert (L/t cane crushed)

Meaning: A warning threshold used to flag when specific fresh-water consumption is higher than the desired operating range. The alert is normally set above the target so that the plant can investigate before performance deteriorates further.

Example: If the target is 150 L/t cane and the alert is set at 170 L/t cane, a daily result above 170 L/t cane should trigger a review of meter readings, crushing rate, leaks, cleaning practices, cooling systems and reuse-water availability. These numbers are illustrative only; each factory should approve its own values.

3. Distillery Fresh Water Target (KL/KL)

Meaning: A specific fresh-water consumption target for the distillery. The unit KL/KL means kilolitres of water per kilolitre of the selected production measure, but the denominator must be explicitly defined in the audit system.

Before entering a value, confirm whether the denominator is KL of ethanol produced, KL of rectified spirit, or another defined product stream. Do not compare figures that use different product bases or plant boundaries.

Calculation template: Distillery fresh-water consumption (KL/KL) = Fresh water used by the defined distillery boundary (KL/day) ÷ production on the agreed basis (KL/day).

Example: If 180 KL/day of fresh water is used and the agreed denominator is 60 KL/day of ethanol, the specific consumption is 180 ÷ 60 = 3 KL/KL. This is a worked example, not a recommended benchmark.

4. Spent Wash Target (KL/KL)

Meaning: A target for spent-wash generation or handling expressed in KL per KL of a defined distillery product. The audit tool must state whether this refers to spent wash generated, treated, concentrated, or sent for a specified disposal/recovery route.

Calculation template, if the field means generation: Spent-wash generation ratio (KL/KL) = Spent wash generated (KL/day) ÷ agreed distillery production (KL/day).

Example: If spent wash generated is 300 KL/day and production on the agreed basis is 60 KL/day, the ratio is 5 KL/KL. This example is only arithmetic; actual ratios depend on feedstock, process configuration and the precise product basis.

5. Sugar Effluent Norm (L/t cane crushed)

Meaning: The effluent quantity criterion entered for the sugar factory, expressed per tonne of cane crushed. The exact boundary is essential: effluent generated, effluent entering the effluent-treatment plant (ETP), or final discharge are different measures.

Formula: Specific effluent quantity (L/t cane) = Effluent flow at the selected measurement point (m^3/day) × 1,000 ÷ cane crushed (t/day).

Example: If the measured effluent flow at the defined point is 250 m^3/day and crushing is 5,000 t/day, the specific quantity is 250 × 1,000 ÷ 5,000 = 50 L/t cane.

Important: Do not treat a value entered in this field as an official regulatory standard unless it has been checked against the current consent conditions and applicable requirements of the relevant pollution-control authority.

6. Recovered Water Share Target (%)

Meaning: The desired share of the plant's defined water demand supplied by recovered or reused water rather than fresh-water intake. The audit must define which streams count as recovered water and which demand streams are included.

Suggested formula:

Recovered water share (%) = Recovered/reused water supplied to the defined uses (m^3/day) ÷ total water supplied to those uses (m^3/day) × 100

Example: If 1,570 m^3/day of recovered water is used and the combined recovered plus fresh water supplied to the same boundary is 4,810 m^3/day, the recovered-water share is 1,570 ÷ 4,810 × 100 ≈ 32.6%. Use this calculation only when the numerator and denominator cover the same plant boundary and time period.

7. Target versus Alert: Practical distinction

Item	Purpose	Action when exceeded
Target	Desired normal operating performance	Review trends and pursue continual improvement
Alert threshold	Early warning that consumption may be abnormal	Verify readings and investigate likely causes
Regulatory limit / consent condition	Formal requirement, where applicable	Follow the applicable compliance and reporting procedure

8. Method for setting reliable targets

1. Define the process boundary and the exact numerator and denominator for every indicator.
2. Check flow-meter calibration, units, time stamps and daily production records.
3. Collect representative data across normal operating conditions and relevant seasons.
4. Calculate the baseline using consistent units and matched time periods.
5. Separate fresh water, recovered water, recirculating water and wastewater flows.
6. Review process needs, equipment capability, water quality requirements and feasible conservation measures.
7. Approve targets and alert thresholds with the plant team; record the owner, approval date and review frequency.
8. Review exceptions, investigate abnormal readings and revise targets only with documented justification.

9. Blank target register

Audit field	Unit	Baseline	Target	Alert / norm	Boundary / notes
Sugar fresh water	L/t cane				
Distillery fresh water	KL/KL				Define production denominator
Spent wash	KL/KL				Define generation/handling boundary
Sugar effluent	L/t cane				Define generation, ETP inlet or discharge
Recovered water share	%				Define included streams and uses

10. Site audit checklist

☐ Are fresh-water, recovered-water and effluent meters identified and functioning?

☐ Are daily cane crushing and distillery production figures recorded on the same time basis as water flows?

☐ Are the distillery product denominator and spent-wash measurement boundary documented?

☐ Is the effluent measurement point clearly marked (generation, ETP inlet or final discharge)?

☐ Are target values distinguished from alerts and statutory/consent requirements?

☐ Are deviations investigated and corrective actions recorded?

☐ Are reuse-water quality and intended use compatible with process and safety requirements?

Important note

The numerical examples in this guide are illustrative calculations, not official benchmarks or regulatory limits. Before adopting targets, verify the definitions used by the Cane Water Audit tool, the plant's approved operating basis, and applicable consent conditions. In particular, confirm the denominator for distillery KL/KL indicators and the measurement boundary for sugar effluent.